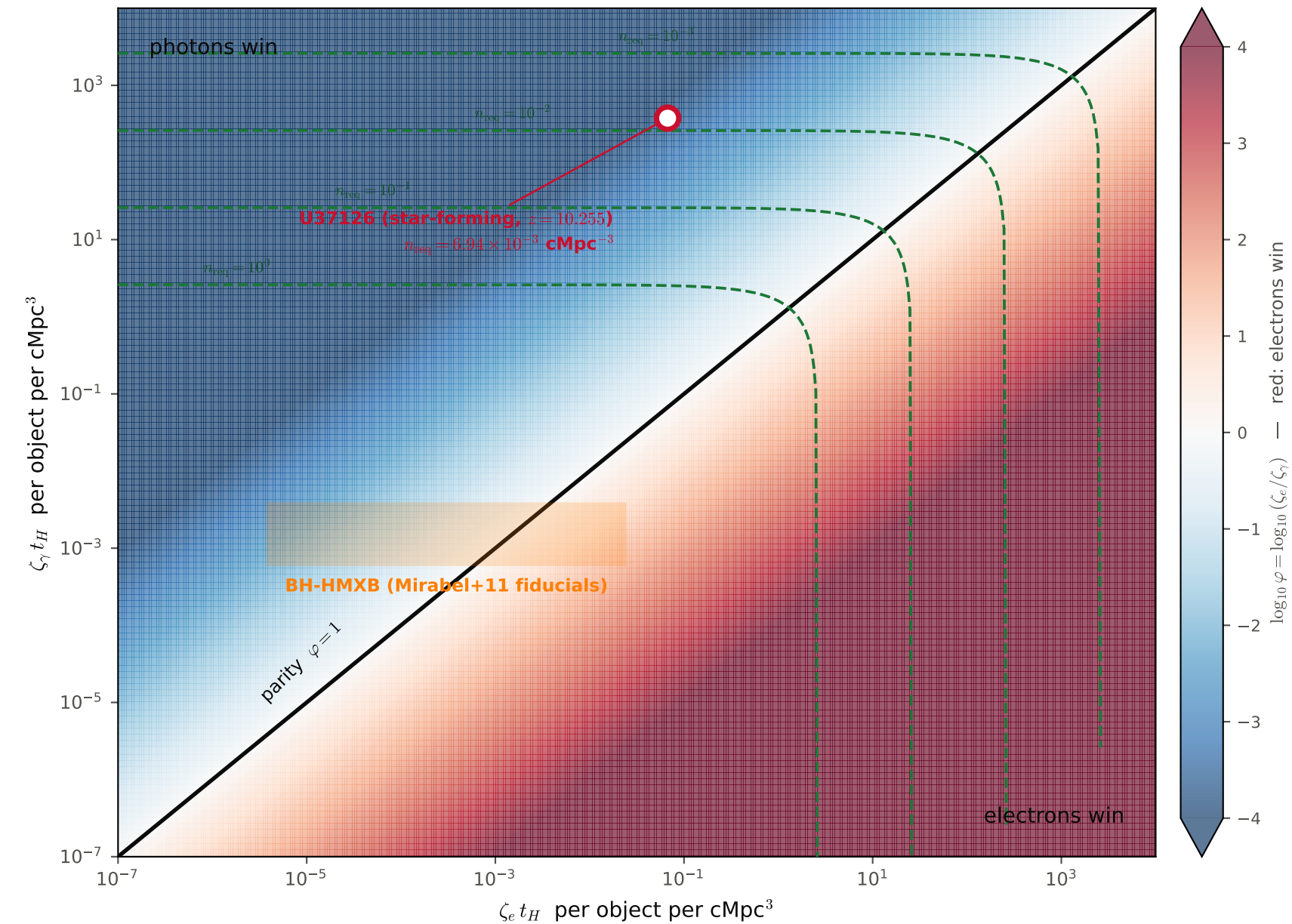


Budget plane for SINGLE OBJECTS at $z=10$: how many would it take?



A point here is ONE object per cMpc^3 , so the dashed anti-diagonals read as the number density that class would need. Compare against an observed luminosity function.

U37126 ($M_{\text{UV}} = -20.1$, Marques-Chaves+26) is EXACT: both channels follow from the chain already in the manuscript. Its n_{req} exceeds the density $\rho_{\text{UV}}/L_{\nu} = 2.77 \times 10^{-4} \text{ cMpc}^{-3}$ that ρ_{UV} implies by the same $\times 25$ the population figure finds --- an independent route to that shortfall.

BH-HMXB uses Mirabel+11 eqs.~(3)--(4): $L_{\text{Edd}} = 1.5 \times 10^{38} (M_{\text{BH}}/M_{\odot})$, $f_{\text{Edd}} = 0.1$, $f_{2-10} = 0.1$, carried to the ionizing band by the Gladstone+09 BHB spectrum. The box spans $M_{\text{BH}} = 5\text{--}30 M_{\odot}$ (OUR choice) and the scanned jet electron fraction; position scales linearly with M_{BH} .

NOT plotted, for want of one specific number each: SS 433 / S26 microquasars; Single SN / SNR; AGN / little red dots. $f_{\text{dep}} = 1$ (A11): every ζ_e is a ceiling.